AI FAQ ASSISTANT - PROJECT REPORT

Phase 2: Requirement Analysis Phase

Software Requirements

• Operating System: Windows 10/11, macOS or Linux
• Node.js: v18 or above
• npm: v9 or above
• Express.js: v4/v5
• MongoDB/Mongoose: Local or Atlas
• Testing Tools: Postman/ThunderClient
• IDE: VS Code or any IDE

Hardware Requirements

• Processor: Intel Core i5 (8th Gen) or Ryzen 5 minimum
• RAM: 8 GB minimum, 16 GB recommended
• Storage: 1 GB free space

Functional Requirements

• User registration with name, email and password
• Authentication and JWT token generation
• User profile access
• FAQ CRUD operations
• Keyword search functionality
• AI-assisted FAQ generation
• Author-based authorization

Non-Functional Requirements

• Security: Passwords hashed with bcrypt, JWT tokens
• Reliability: Centralized error handling
• Maintainability: Modular code structure
• Scalability: Stateless JWT sessions
• Performance: Sub-second response times
• Portability: Cross-platform compatibility

Feasibility Study

• Technical: Open-source tools with good documentation and team expertise
• Economic: Free software stack with free Gemini API quota
• Operational: Simple to run and easy for clients to consume

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